

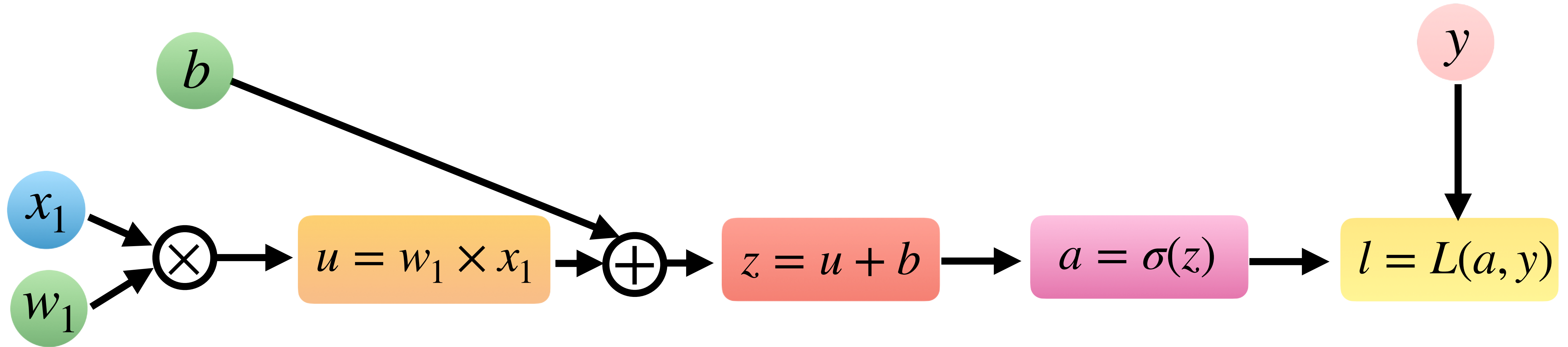
3.3

Model Training with Stochastic Gradient Descent

Part 3: Minimizing the Loss Using Gradient Descent

Sebastian Raschka and the PyTorch Lightning Team

Chain Rule

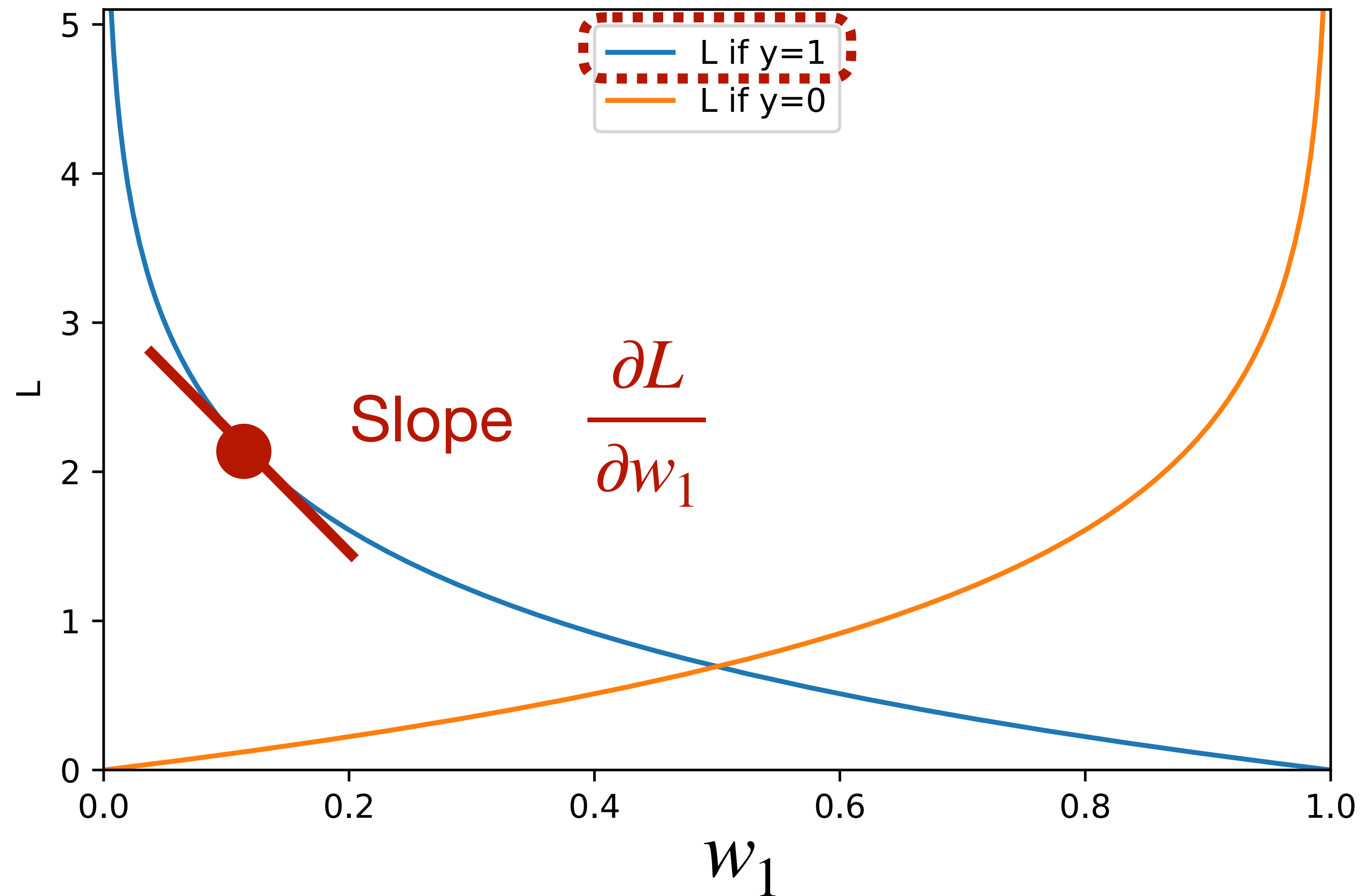


$$\frac{\partial L}{\partial w_1} = \frac{\partial u}{\partial w_1} \times \frac{\partial z}{\partial u} \times \frac{da}{dz} \times \frac{\partial L}{\partial a}$$

$$\frac{\partial L}{\partial b} = \frac{\partial z}{\partial b} \times \frac{da}{dz} \times \frac{\partial L}{\partial a}$$

$$\nabla L = \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial b} \end{bmatrix}$$

Changing parameters using the negative slope



The Gradient Descent Algorithm

The Gradient Descent Algorithm

For each training epoch:

`overall_loss` $L = 0$

The Gradient Descent Algorithm

For each training epoch:

overall_loss $L = 0$

For each training example:

output = $\sigma(z)$

overall_loss $L = \text{overall_loss } L + l(\text{output}, \text{label})$

The Gradient Descent Algorithm

For each training epoch:

overall_loss $L = 0$

For each training example:

output = $\sigma(z)$

overall_loss $L = \text{overall_loss } L + l(\text{output}, \text{label})$

overall_loss $L = \text{overall_loss } L / n$

The Gradient Descent Algorithm

For each training epoch:

overall_loss $L = 0$

For each training example:

output = $\sigma(z)$

overall_loss $L = \text{overall_loss } L + l(\text{output}, \text{label})$

overall_loss $L = \text{overall_loss } L / n$

compute $\nabla L_{\mathbf{w}}$ and ∇L_b

The Gradient Descent Algorithm

For each training epoch:

overall_loss $L = 0$

For each training example:

output = $\sigma(z)$

overall_loss $L = \text{overall_loss } L + l(\text{output}, \text{label})$

overall_loss $L = \text{overall_loss } L / n$

compute $\nabla L_{\mathbf{w}}$ and ∇L_b

$\mathbf{w} = \mathbf{w} + \alpha \nabla L_{\mathbf{w}}$

$b = b + \alpha \nabla L_b$

The Gradient Descent Algorithm

For each training epoch:

`overall_loss` $L = 0$

Computes the loss over the whole epoch

For each training example:

`output` = $\sigma(z)$

`overall_loss` $L = \text{overall_loss } L + l(\text{output}, \text{label})$

`overall_loss` $L = \text{overall_loss } L / n$

compute $\nabla L_{\mathbf{w}}$ and ∇L_b

$\mathbf{w} = \mathbf{w} + \alpha \nabla L_{\mathbf{w}}$

$b = b + \alpha \nabla L_b$

The Gradient Descent Algorithm

For each training epoch:

`overall_loss` $L = 0$

For each training example:

`output` = $\sigma(z)$

`overall_loss` $L = \text{overall_loss } L + l(\text{output}, \text{label})$

`overall_loss` $L = \text{overall_loss } L / n$

compute $\nabla L_{\mathbf{w}}$ and ∇L_b

$\mathbf{w} = \mathbf{w} + \alpha \nabla L_{\mathbf{w}}$

$b = b + \alpha \nabla L_b$

Update parameters once per epoch

The **Stochastic** Gradient Descent Algorithm